

Pavan Kumar Gannaju

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PROFESSIONAL SUMMARY

Backend / Platform Engineer focused on Java, Go, PostgreSQL, Kafka, Kubernetes, Azure, and CI-backed verification. Built evidence-backed systems including a cloud-validated double-entry ledger, Kubernetes admission controller, secure desktop-agent prototype, and Raft-backed key-value store. Strong focus on correctness, security boundaries, failure handling, and reproducible tests.

SKILLS

Languages: Java, Go, Python, Rust, SQL, JavaScript/TypeScript

Backend: Spring Boot, FastAPI, REST APIs, gRPC, JWT/OAuth2, OpenAPI

Data & Messaging: PostgreSQL, Redis, Kafka, SQLite

Cloud & Infrastructure: Azure, Docker, Kubernetes, kind, Helm, Kustomize, GitHub Actions, Linux

Security & Supply Chain: cosign, Trivy, SBOM, safetensors validation, admission control, prompt-injection defense

Testing & Verification: Testcontainers, envtest, fuzzing, property-based testing, k6, Porcupine, fault injection, QEMU

Core CS: Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Distributed Systems

EXPERIENCE

AI Intern with Research Team Lead Responsibilities, Prodigal AI Technologies Pvt. Ltd., Delhi *Mar 2025 – Nov 2025 / Remote*

- Led a 5-intern R&D team developing generative-AI security and trustworthy-AI systems, coordinating sprint delivery, reviewing implementation decisions, and establishing research and evaluation protocols.
- Engineered modular Python backend workflows for internal AI systems, integrating APIs, input-validation logic, and dedicated test scenarios.
- Designed and ran empirical benchmarking as part of the team's research methodology, analyzing experimental results and contributing findings to research manuscripts on voice-cloning input control and defense-in-depth against prompt injection, data poisoning, and adversarial manipulation.

Gen AI / LLM Intern, Digital Nexus AI, Jersey City, NJ, USA

May 2025 – Sep 2025 / Remote, Part-Time

- Engineered Python/FastAPI REST APIs and backend services supporting service-to-service communication across AI workflow components.
- Implemented request-processing pipelines, input validation, and integration utilities for AI-enabled backend workflows.
- Investigated backend failures using application logs and request traces, reproduced defects, and added unit and functional tests before release handoff.

PROJECTS

LedgerLine — Double-Entry Financial Ledger / *Java 21 · Spring Boot · PostgreSQL · Kafka · Docker · Azure · GitHub Actions · k6 · Testcontainers* | [Github](#)

- Built a double-entry ledger where PostgreSQL enforces zero-sum postings and non-negative balances, with idempotent transfers, deterministic lock ordering, a transactional outbox feeding Kafka projections, and JWT-secured APIs.
- Benchmarked concurrency strategies with k6: ordered pessimistic locking sustained 258.9 req/s under high-contention local Docker Compose workload, while optimistic and serializable strategies degraded under retry/abort exhaustion.
- Validated on Azure using ACR, Azure PostgreSQL Flexible Server, and VM-based Docker; smoke checks, k6 correctness tests, PostgreSQL invariants, and resource-group teardown all passed. CI runs Testcontainers integration tests, failure-path tests, SBOM generation, and Trivy scans.

QuorumKV — Replicated Key-Value Store and Fault-Injection Lab / *Go · gRPC · HashiCorp Raft · WAL · Porcupine · GitHub Actions* | [Github](#)

- Built a replicated key-value service on HashiCorp Raft with a custom CRC-framed, fsync-before-ack WAL, gRPC API, and multi-process cluster tests.
- Verified crash recovery across 100 randomized process-kill trials: acknowledged writes survived and no writes were invented.
- Built a TCP fault-injection harness for partitions, delays, rolling failures, and double-leader scenarios; checked five recorded histories with Porcupine linearizability tests. Consensus is provided by HashiCorp Raft; I built the service behavior, WAL durability, gRPC API, fault harness, measurements, and verification.

ModelGate — Kubernetes Admission Control for ML Workloads / *Go · Kubernetes · controller-runtime · cosign · safetensors · Helm · Kustomize · envtest* | [Github](#)

- Built a Kubernetes admission webhook that admits only cosign-signed container images and hash-verified safetensors model artifacts while denying pickle-like payloads, privileged containers, host access, and unsafe workload updates.
- Found and fixed two real admission bypasses: post-admission image swaps via kubectl set image and ephemeral-container injection.
- Validated with unit tests, envtest, fuzzing, real cosign sign/verify integration, kind-cluster smoke tests, Helm/Kustomize install paths, and Azure ACR evidence.

SentinAL — Secure Desktop-Agent Orchestration Prototype / *Python · FastAPI · Windows Automation · SQLite · Ollama* / [Github](#)

- Led a 5-member B.Tech thesis team building a Windows desktop-agent prototype that treats the LLM as untrusted and enforces allowlists, sandboxing, confirmation gates, and postcondition checks outside the model.
- Recorded 96.7% end-to-end success across 120 evaluated tasks using independent Windows OS-state checks; blocked 66/66 adversarial cases in the recorded fuzz suite.
- Scope is intentionally bounded: Windows-only, with 6 of 19 intents still lacking independent postcondition checks.

EDUCATION

M.Sc. Research in Computer and Systems Engineering — Incoming Student | Technische Universität Ilmenau, Germany · April 2027

B.Tech in Computer Science and Engineering (AI/ML) — Jawaharlal Nehru Technological University Hyderabad | Graduated April 2026 | CGPA: 8.01/10.0

CERTIFICATIONS

- Google Cybersecurity Professional Certificate
- Microsoft Cybersecurity & OS Fundamentals
- SAP Code Unnati Advanced Training Program

CODING & LEADERSHIP

- Solved 530+ LeetCode problems, including 86 Hard; solved 249 Codeforces problems, including 218 in 2026.
- Led 5-member development/research teams during internship and B.Tech thesis work; organized and mentored university hackathons with 50+ participants.